

PART 1 – Emotional Interaction (Updated 2024–25)

This builds on your earlier Emotional Interaction lecture (Week 12) but focuses on **new research trends**.

1. What is “Emotional Interaction” (updated view)?

Your slide says: emotional interaction is about **systems that sense, interpret, or respond to human emotions**.

But new points:

- It **goes beyond** old “affective computing” (just recognizing emotions).
- It also includes:
 - **Ethics** – is this fair / manipulative?
 - **Culture** – different cultures show emotions differently.
 - **User agency** – does the user stay in control of their feelings?

Trend: shift from “**control the user’s emotion**” → “**support the user’s emotion**”.

👉 MEMORIZE:

- Emotional interaction = sense + interpret + respond to emotion + consider ethics, culture, user control.

2. Trend 1 – Culture-Aware Emotional Intelligence

Old emotional AI:

- Uses “universal” emotion models (e.g., smile = happy, frown = sad)
- Problem: These are often **Western-biased**. Different cultures:
 - Show emotions differently

- Interpret faces differently

New research: **Culture-aware emotion models.**

Your slide mentions **Affective–CARA framework**:

- It uses **knowledge graphs** (structured knowledge) to:
 - Include cultural context
 - Reduce misinterpretation
- For example:
 - A “neutral” face in one culture might be seen as “cold” in another.
 - Certain gestures (eye contact, hand signs) differ culturally.

Design implication:

- Don’t assume **one expression = same meaning everywhere.**
- Emotion systems should adapt to **local culture.**

👉 **MEMORIZE:**

- Name “**Affective–CARA**” as an example of **culture-aware emotional AI** that reduces cultural bias.

3. Trend 2 – Emotion *Support* vs Emotion *Regulation*

Old approach (slightly toxic):

- System tries to **regulate** your emotion:
 - “Calm you down”
 - “Make you happy”

- “Fix” your sadness

New critique:

- This can be **controlling** and may ignore user’s real needs.

New approach: **Support, not control.**

- Systems should:
 - Help you **express** emotions
 - Help you **understand** them
 - Give **space and tools**, not force a desired mood
- Example:
 - A journaling app that lets you log anger and reflect, instead of saying “Stop being angry, here’s a cat GIF.”

Design implication:

- Ask: “Am I helping the user, or **forcing** an emotion on them?”

👉 **MEMORIZE sentence:**

“New emotional systems focus on **supporting emotional expression and agency**, not **regulating or fixing** emotions.”

4. Trend 3 – Emotional Output Ethics (R-CAGE Model)

Now we’re not just reading emotions, we’re **outputting** them (chatbots, agents).

The slide mentions **R-CAGE model** (for emotional output ethics).

R-CAGE (you don’t need the full expansion; just what it’s about):

- It guides **how AI should express emotion**, including:

- **Tone** – gentle, harsh, neutral?
- **Pacing** – how often? too spammy?
- **Intensity** – mild empathy vs dramatic oversharing
- Goal:
 - Avoid **manipulating** users' emotions
 - Keep emotional responses **ethical and respectful**

Example:

- Bad: A mental health bot that acts overly dramatic to keep you engaged so you stay hooked = manipulation.
- Good: Bot that responds calmly, acknowledges feelings, but **does not push** you into buying products or changing beliefs for its own benefit.

👉 MEMORIZE:

- **R-CAGE** = model for **ethical emotional output** (tone, pacing, intensity, no manipulation).

5. Trend 4 – Emotion-Aware Companions (e.g., Livia)

Your slide gives example: **Livia**, an AR-based emotionally adaptive companion.

Features:

- Uses **multimodal emotion detection**:
 - Facial expressions
 - Voice tone
 - Text

- Maybe physiological signals
- Uses **memory compression**:
 - Doesn't store everything raw
 - Keeps a compact "summary" of your history to respect privacy a bit more and stay efficient.

Applications:

- **Wellness** – checking in on your mood.
- **Loneliness** – giving emotional support.
- **Social support** – like a companion that remembers your patterns.

Key concern:

- Needs **ethical guidelines**: not a manipulative "fake friend".

👉 **MEMORIZE:**

- Example name "**Livia**" as an **AR emotion-aware companion** for wellness and loneliness support.

6. Trend 5 – Immersive Empathic Computing (VR/AR side of emotions)

This trend overlaps with your VR deck.

Empathic computing:

- Systems that create **strong emotional presence** so you:
 - Feel what others feel
 - Experience situations deeply (e.g., VR empathy training)

Key points:

- **VR dominates** empathic computing research.
 - High emotional presence → you feel “inside” the situation.
- **AR/MR** are emerging for empathic scenarios too.
- Often uses **dimensional models of emotion**:
 - **Valence** (positive ↔ negative mood)
 - **Arousal** (calm ↔ excited)

Strengths:

- Very strong **emotional impact**.
- Good for training empathy (e.g., experiencing life as a patient, refugee, disabled person).

Challenges:

- **Motion sickness**, comfort.
- **Ethical concerns**:
 - Too intense?
 - Trauma?
 - Manipulation?

7. Trend 6 – Advances in Affective Sensing

Affective sensing = sensing emotion from signals.

New direction:

- **Integrated** sensing:
 - Physiological (heart rate, GSR, breathing)

- Behavioral (face, voice, typing, phone use)
- Real-time adaptation:
 - Interface adapts based on stress or boredom levels.

Applications:

- **Wearables** – watches detecting stress.
- **Adaptive learning platforms** – change difficulty when you seem frustrated.
- **Wellbeing tools** – breathing reminders, posture nudges.

Design concerns:

- Must be **transparent**:
 - Tell user what's being sensed and why.
 - Respect **privacy & consent**.
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8. Design Implications – Emotional HCI 2025

Your slide ends with core design implications.

Summarize:

1. **Respect user agency**
 - Don't over-control emotions.
2. **Cultural sensitivity**
 - Avoid one-size-fits-all emotion models.
3. **Transparency**
 - Be clear about:

- What you sense
- How you use emotional data
- How you express emotion

👉 **MEMORIZE 3 bullets:** *Agency, Culture, Transparency.*

PART 2 – Ubiquitous Computing, VR, AR & Visualisation

Now the second file: *ubiquitous computing and augmented realities + VR/AR.*

1. Big picture: Ubiquitous computing vs VR/AR

Slide summary:

- **Ubiquitous computing (ubicom):**

Filling the **real world** with computers.
- **Virtual & augmented reality:**

Making the **real world inside a computer**
Or **mixing real + virtual**.

Weiser's famous quote:

“The most profound technologies are those that **disappear**.”

Meaning:

- When tech is truly embedded in life, you **stop noticing** it (like electricity, Wi-Fi).
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2. Ubiquitous Computing – definition & implications

Definition:

Any computing tech that lets humans interact **away from a single workstation** (i.e., not just desktop).

Implications:

- Technology now defines:
 - The **interactive experience**
 - **Applications** (smart homes, wearables, public displays)
 - **Theories of interaction** (activity theory, situated action).
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3. Scales of devices (Weiser's inch / foot / yard)

Weiser proposed device scales based on **physical size**.

3.1 Inch-scale devices

Examples:

- PDAs (old), small recorders
- Smartphones
- Smartwatches, sensors

Characteristics:

- Owned by **individuals**
- Many per person
- Can communicate with each other and environment.

3.2 Foot-scale devices

Examples:

- Notebooks / laptops
- Tablets
- Digital paper devices

Characteristics:

- Individuals may own several
- Not always on them.

3.3 Yard-scale devices

Examples:

- Electronic whiteboards
- Plasma displays
- Smart bulletin boards

Characteristics:

- Owned by **buildings or institutions**
- Used by **many people**.

👉 **MEMORIZE:** inch = phones/watches; foot = laptops/tablets; yard = shared large displays.

4. Defining the Interaction Experience in Ubicomp

4.1 Implicit input

- **Sensor-based** input instead of explicit keyboard/mouse.
- Recognition tech: motion sensors, proximity, cameras, microphones.

Pros:

- More natural (system is “aware”).

Cons:

- Recognizers are **not perfect** → ambiguity, errors.

4.2 Multi-scale & distributed output

- Screens of many sizes:
 - Very small (watch)
 - Very large (wall screens)
- Distributed in space, but **coordinated**.

4.3 Ambient & peripheral displays

- More than just classic screens.
- Use physical environment to **signal info in the background**.

Examples:

- **Dangling String** – string that moves faster when network traffic is high.
- **Water Lamp** – lights/water movement shows status.

These are meant to be:

- **Ambient** – part of the environment.
- **Peripheral** – not demanding full attention.

5. Merging physical & digital worlds

Goal: **remove barrier** between physical actions and digital effects.

- Actions on physical objects have **digital meaning** (and vice versa).
- Examples in slides:
 - “Digital desk” – projections on physical desk.
 - “Augmented calendar” – physical page with digital overlays.
- Digital, smart, and augmented objects.

Application themes:

1. **Context-aware computing**
 - Sense environment to ease interaction (location, light, motion).
2. **Automated capture and access**
 - Capture live experiences (video, sensors) for future replay.
3. **Toward continuous interaction**
 - No clear begin-end; everyday life is continuously supported.

6. New theories used in Ubicomp HCI

The slide lists theories that become more important in ubicomp:

- **Knowledge in the world**
 - Design uses the environment as external memory (labels, physical cues).
- **Activity theory**
 - Focuses on how goals and actions **adjust fluidly** to the physical world.
- **Situated action & distributed cognition**
 - Emphasizes **improvisation** and opportunistic behavior, rather than strict planned steps.

- **Ethnography**
 - Deep study of people in their natural environments.

The idea: ubicomp is messy & real-life → we need **real-world observation theories**, not just neat lab tasks.

7. Evaluation challenges in Ubicomp

Slides ask: how do we evaluate these systems?

Problems:

- Activities are **not so task-centric** (no neat “step 1–2–3”).
- Tech is **new**, so:
 - Hard to get long-term authentic use.
- **Success metrics** differ:
 - Not just efficiency; might be:
 - **Playfulness**
 - **Non-distraction**
 - **Enjoyment**

Examples:

- **Ambient Wood:**
 - Real forest augmented with sensors, sound, location tracking.
- **City – shared experience:**
 - People in VR, on web, and physically in a museum share experiences, talk via microphones, see each other virtually.

8. VR vs AR – basics

8.1 VR – Virtual Reality

Definition:

VR: technology + experience where a computer simulates the **real or imaginary world**, and the user is immersed inside it.

Technology:

- **Headsets:**
 - One small LED screen per eye, slightly different image → **3D effect**.
- **DataGlove:**
 - Lycra glove with sensors to track hand positions.
- **Eye-gaze:**
 - Track where you look.
- **Whole-body tracking:**
 - Position, walking, gestures.

Experience:

- Mainly **visual**, but also sound, haptics, gesture.
- Aim: **immersion, engagement, interaction**.

Uses:

- Simulate life-like but:
 - Too dangerous (firefighting, surgery)
 - Too expensive

- Too impossible (moon walk, micro-scale molecules)

8.2 Desktop VR

Headsets can be:

- Expensive, uncomfortable.
Solution: **Desktop VR**:
- Use normal monitor + PC.
- Cheaper, more accessible.

On the web:

- **VRML** (Virtual Reality Markup Language) – old standard to define 3D scenes for the web.

Command & control:

- Scenes projected onto walls, with real controls and hydraulic rams (e.g., flight simulators).

8.3 AR – Augmented Reality

Definition:

AR = overlaying **virtual information** on the **real world**.

Examples:

- Aircraft **head-up display** (HUD) – projecting info on cockpit glass.
- Semi-transparent goggles that project info on a desktop.

Types of information:

- **Unrelated:**

- Example: reading email on glasses while walking.
- **Related:**
 - Virtual objects that interact with real world (e.g., arrows pointing to real parts).

Key issue: **registration**

- **Aligning virtual and real correctly:**
 - If arrow is off by a bit, user may be misled.

Applications:

- **Maintenance:**
 - Overlay instructions/schematics on real machines (photocopiers, aircraft wiring).
 - Sometimes registration is too hard → use schematic instead.
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9. Applications of VR & AR

VR Applications

- **Simulation & training:**
 - Flight simulators, ships, military.
- **Games:**
 - Immersive gaming.
- **VR holidays:**
 - Rainforest, safari, skiing, moon walk.
- **Medical:**
 - Practice surgery using 3D scans and force feedback.

- **Phobia treatment:**
 - Virtual lifts, spiders, etc., in controlled environment.

AR Applications

- Maintenance, repair, complex wiring.
 - Instruction overlays.
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10. Information & Data Visualisation

Big idea:

Use 2D/3D visuals (sometimes VR) to understand complex data.

Types:

- **Scientific/technical data** (continuous values)
- **Structured information** (hierarchies, networks)
- **Temporal data** (events over time)

10.1 Scientific & technical data

- Real space is 3D → use 3D space to visualise.
- Example: **virtual wind tunnel**:
 - Fluid dynamics simulate air flow over objects.
 - Virtual bubbles show flow direction → cheaper & no physical tunnel.

For 2D space:

- Geographic data: map with values projected as heights/colours.

Visualisation for no “real” dimensions:

- Histograms, scatter plots, pie charts, etc.

10.2 Structured information

- Hierarchies: file trees, organisation charts.
- Networks/graphs: web pages, links.

Problems:

- Wide hierarchies → large 2D charts.

Solution:

- Use **3D** representations like **Cone Trees** (Xerox):
 - Levels become rings in 3D.
 - Overlap OK because you can rotate.

Networks:

- Planar vs non-planar graphs.

10.3 Time & interactivity

Visualising time:

- Map time to **space**:
 - Sales graphs, Gantt charts, timelines.
- Map time to **time**:
 - Replay events at different speeds.
- Map **space** to **time**:
 - Visible Human Project: scroll through slices of a body.

Interactivity:

- Let user adjust what they see:
 - Change thresholds, filters, zoom.
 - Example: **Influence Explorer** (for exploring complex data).
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11. Summary: Between Two Worlds

Final slide:

- **Ubiquitous computing**: computers in the real world.
- **Virtual reality & visualisation**: real world **represented** inside computers.
- **Augmented reality & ambient displays**: physical and digital **intermingled**.

Maturity:

- VR & visualisation: already fairly common.
 - AR & ubicomp: rapidly growing.
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PRACTICE – Week 13 Questions

A. 5 MCQs (with answers + explanations)

MCQ 1

In recent emotional interaction research, what is the main **shift** regarding how systems handle user emotions?

- A. From supporting emotions to strictly regulating them
- B. From recognizing emotions to ignoring them
- C. From regulating/fixing emotions to supporting users' emotional expression and agency
- D. From using multimodal sensing to face-only sensing

Answer: C

Explanation:

New research criticizes systems that try to “fix” emotions and instead focuses on **supporting emotional expression and user agency**, not controlling how users should feel.

MCQ 2

The **Affective–CARA** framework is mainly introduced to:

- A. Increase the intensity of emotional feedback
- B. Reduce cultural bias in emotion interpretation
- C. Replace physiological sensing
- D. Regulate user emotions using fixed rules

Answer: B

Explanation:

Affective–CARA uses knowledge graphs to build **culture-aware emotional intelligence**, aiming to avoid one-size-fits-all, Western-biased emotion models.

MCQ 3

Which device scale in Weiser’s model typically includes **smartphones and smartwatches**?

- A. Yard-scale
- B. Inch-scale
- C. Foot-scale
- D. Nano-scale

Answer: B

Explanation:

Inch-scale devices are small personal devices like PDAs, smartphones, and smartwatches, owned individually and often carried everywhere.

MCQ 4

Which of the following is the best description of **augmented reality (AR)** from your slides?

- A. A fully simulated world replacing reality
- B. A technology that overlays virtual information on the real world
- C. A desktop-based 3D modelling tool
- D. A method for coding VRML files

Answer: B

Explanation:

AR overlays virtual images or information on the physical environment, e.g., head-up displays or semi-transparent goggles showing data on top of real objects.

MCQ 5

What is one of the main **evaluation challenges** in ubiquitous computing environments?

- A. Users refuse to use any sensors
- B. Activities are not task-centric and “success” may be about playfulness or non-distraction rather than efficiency
- C. Devices cannot be networked together
- D. VR headsets cannot be used in ubicomp settings

Answer: B

Explanation:

Ubicomp activities are often not well-defined tasks, technologies are new, and metrics of success may be **playfulness or non-distraction**, not traditional efficiency, making evaluation harder.

B. 5 Short Answer Questions (with model answers)

SAQ 1

What are the three key design implications for emotion-aware systems mentioned in your Week 13 slides?

Model answer:

The slides emphasize that emotion-aware systems should:

1. **Respect user agency** – support users’ own emotional choices rather than controlling them.

2. Include **cultural sensitivity** – avoid universal, biased emotion interpretations by adapting to cultural context.
 3. Provide **transparent emotion sensing and output policies** – clearly inform users what is sensed, how it is interpreted, and how emotional outputs are generated.
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SAQ 2

Define ubiquitous computing and give one example of an inch-scale and one example of a yard-scale device.

Model answer:

Ubiquitous computing is any computing technology that allows human interaction **away from a single workstation**, embedding computers into everyday environments. An **inch-scale** example is a smartphone or smartwatch, which individuals carry. A **yard-scale** example is an electronic whiteboard or smart bulletin board, owned by a building and shared by many people.

SAQ 3

What is “implicit input” in ubiquitous computing, and why does it introduce ambiguity?

Model answer:

Implicit input is **sensor-based input** where systems infer actions or context from things like motion, location, or presence, rather than explicit commands like keyboard input. It introduces ambiguity because recognition technologies (e.g., motion detectors, computer vision) are not perfect, so the system may misinterpret user actions or context.

SAQ 4

Explain the difference between VR and AR using examples from the slides.

Model answer:

VR (Virtual Reality) immerses the user in a **fully simulated world**, often using headsets and devices like DataGloves to interact, e.g., flight simulators or VR training for surgery. AR (Augmented Reality) **overlays virtual information** onto the real world, such as aircraft head-up displays or semi-transparent goggles that show maintenance instructions directly on top of real machinery.

SAQ 5

What is the purpose of using visualisation techniques like cone trees or virtual wind tunnels?

Model answer:

Visualisation techniques like **cone trees** help represent complex hierarchies in 3D so that large, wide structures are easier to explore, while **virtual wind tunnels** use 3D visualisations of airflow over objects to study fluid dynamics cheaply and without disturbing real air flow. In both cases, the goal is to make complex or invisible data **more understandable and explorable**.

C. Case Study (with full answer)

Scenario

You are designing a **VR-based empathy training experience** for medical students to help them understand what it's like to be a hospital patient.

The experience uses:

- A VR headset
- Emotional sensing (heart rate, facial expression)
- A virtual companion (agent) inside the VR world

Your tasks:

1. Using **Week 13 emotional interaction trends**, explain two ways you should be careful about **ethics and user agency** in this system.
 2. Using **ubiquitous/VR concepts**, explain why VR is a good medium for this empathy training and mention one **challenge**.
 3. Suggest one way to evaluate this system, given the **ubicom evaluation challenges** discussed in the slides.
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Model Answer

1. Ethics and user agency in emotional interaction

First, from the trend “**Emotion support vs regulation**”, the system should **support** medical students in exploring emotions, not **force** them to feel guilty or overwhelmed. For example, the virtual companion can gently reflect what the patient might feel (“You look tense; this must feel overwhelming”) instead of aggressively pushing guilt (“You don’t care about patients enough”). This respects their **emotional agency**.

Second, in line with **R-CAGE emotional output ethics**, the virtual companion’s **tone, pacing, and intensity** must be carefully controlled. Emotional feedback should not be manipulative or excessively intense, to avoid psychological harm; the system should be transparent about using heart-rate and facial expressions for adaptation. This addresses ethical emotional output and user consent.

2. Why VR is suitable + one challenge

VR is suitable because it offers **high immersion and emotional presence**; students can experience a patient’s viewpoint—lying in a bed, hearing monitors, seeing staff from below—creating a strong empathic effect that is hard to get from textbooks. This aligns with **immersive empathic computing**, where VR is a dominant medium for emotional presence.

However, VR also has challenges: some students may experience **motion sickness or discomfort**, and the intense emotional content could be ethically tricky if not paced properly. The system must provide breaks and allow users to **exit** when needed.

3. Evaluation approach given ubicomp challenges

Because empathy and emotional learning are **not simple task metrics** like “time to complete a form”, evaluation should focus on **qualitative and experiential** outcomes, not just efficiency. For example:

- Conduct pre- and post-questionnaires on empathy and attitudes.
- Observe students in **real-world interactions** with standardized patients before and after VR exposure.
- Use interviews to understand how the experience affected their thinking.

This matches the ubicomp challenge that success metrics may be about **playfulness, reflection, or non-distraction**, not just speed or accuracy, so evaluation must include **rich, contextual data** rather than only traditional performance measures.